رواد مصر الرقمية

وزارة الاتصالات
وتكنولوجيا المعلومات

# Loan Risk Detection Project

## Team Members :

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## Project Description:

The objective of this project is to develop a machine learning system capable of predicting loan default risk for financial institutions. Loan risk detection is a critical task in the finance sector, as it helps organizations evaluate the likelihood of a borrower failing to repay a loan, enabling better decision-making and reducing financial losses.

This project focuses on analyzing borrower data such as demographics, income, credit history, employment status, and loan details to identify patterns associated with default risk.

The workflow includes data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model development, and evaluation. Multiple machine learning algorithms will be applied and compared to select the most effective model.

## Dataset :

From Kaggle dataset

## About Dataset :

Financial loan services are leveraged by companies across many industries, from big banks to financial institutions to government loans. One of the primary objectives of companies with financial loan services is to decrease payment defaults and ensure that individuals are paying back their loans as expected. In order to do this efficiently and systematically, many companies employ machine learning to predict which individuals are at the highest risk of defaulting on their loans, so that proper interventions can be effectively deployed to the right audience.

## Description :

This dataset has been taken from Coursera's Loan Default Prediction Challenge and will provide you the opportunity to tackle one of the most industry-relevant machine learning problems with a unique dataset that will put your modeling skills to the test. The dataset contains 255,347 rows and 18 columns in total.

# Methodology:

## 1- Data Collection & Preprocessing

- Handle missing values
- Encode categorical variables
- Normalize numerical features

## 2- Exploratory Data Analysis (EDA)

- Identify trends and correlations
- Detect key factors influencing loan default

## 3- Feature Engineering

- Create new meaningful features
- Select important variables

## 4- Model Development

- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machines (SVM)

## 5- Model Evaluation

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

## 6- Deployment

- Simple web interface or dashboard for real-time predictions

## Expected Deliverables:

- Cleaned and processed dataset
- Trained machine learning models
- Interactive dashboard or web interface
- Final report with insights and recommendations

## Tools and Technologies:

- Python
- scikit-learn
- pandas
- NumPy
- Power BI / Streamlit (for visualization and deployment)

## Project Management & Execution Plan

### Project Timeline (8 Weeks):

- **Week 1:** Understanding loan risk & project requirements
- **Week 2–3:** Data collection & preprocessing
- **Week 4–5:** EDA & feature engineering
- **Week 6–7:** Model development & evaluation
- **Week 8:** Deployment + report & presentation

## Milestones:

- Dataset prepared
- Insights extracted from EDA
- Models trained and evaluated
- Final model selected
- Dashboard/interface completed
- Final report delivered

## Task Assignment & Team Roles :

1. **Project Manager / Business Analyst**
    - Define objectives and KPIs
    - Manage workflow and coordination
2. **Data Engineer**
    - Data collection and preprocessing
3. **Data Analyst**
    - Perform EDA and extract insights
4. **Machine Learning Engineer**
    - Build, train, and evaluate models
5. **Visualization & Deployment Specialist**
    - Create dashboard / deploy model
    - Prepare final presentation

## Risk Assessment & Mitigation Plan:

| Risk | Impact | Mitigation |
|---|---|---|
| Poor data quality | Low accuracy | Data cleaning & preprocessing |
| Imbalanced data | Biased predictions | Use SMOTE / resampling |
| Overfitting | Poor generalization | Cross-validation & regularization |
| Time constraints | Delay | Proper task distribution |
| Deployment issues | System failure | Early testing & simple tools |

## KPIs (Key Performance Indicators):

**Model Performance:**

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

**System Performance:**

- Response time
- Reliability

**Business Impact:**

- Improved risk detection
- Reduction in loan defaults

**Usability:**

- Ease of use
- Clarity of insights

## Project Outcome:

This project will deliver a practical solution that enables financial institutions to **identify high-risk applicants,** improve lending decisions, and minimize potential losses through data-driven insights.